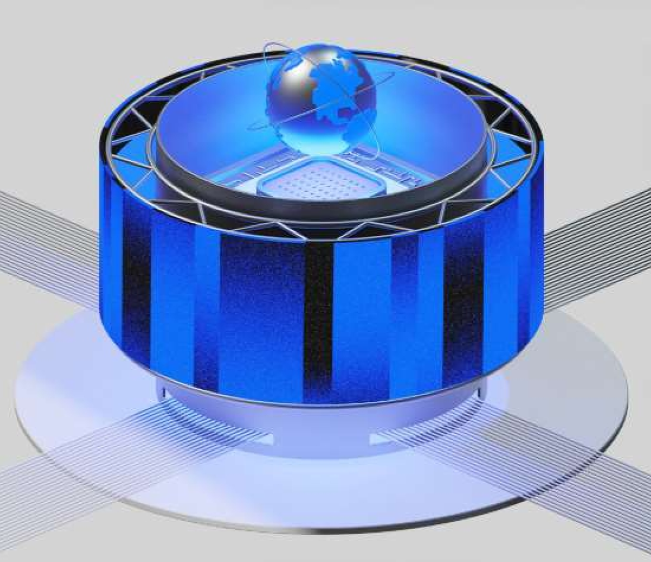
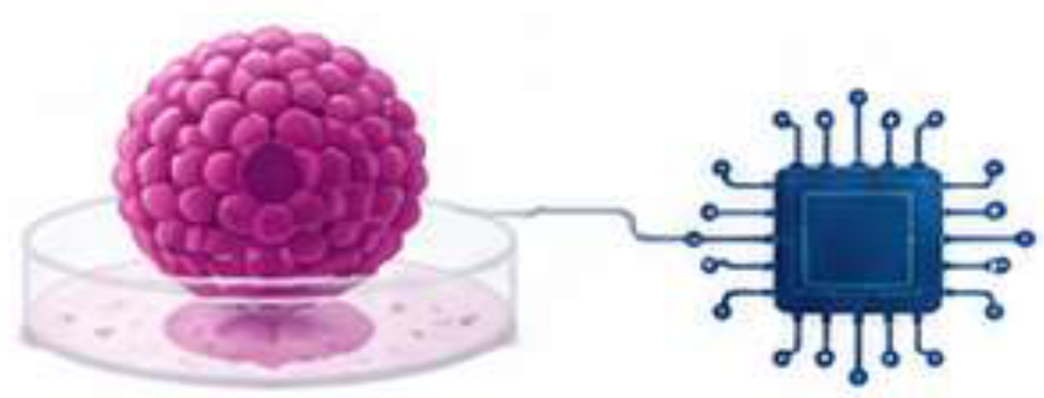


Global Conference on AI, Security and Ethics 2026

Post-Silicon AI and Global Security: Governing Organoid Intelligence



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RESEARCH QUESTION

How can the international community develop an inclusive, effective and anticipatory governance framework to manage the security, ethical and dual-use risks of organoid intelligence systems while enabling responsible innovation?

WHY IMPORTANT FOR SECURITY?

- Organoid AI technologies could transform the technological substrate of AI, creating new strategic advantages and risks.
- Dual-use potential may enable advanced decision-making, autonomous systems and strategic simulations with unpredictable behaviours.
- Bio-digital nature introduces novel biosecurity, ethical and control challenges.
- Fragmented governance and regulatory gaps may lead to misuse, arms race dynamics and regional or global instability.
- Anticipatory governance is essential to safeguard international peace, security and human values.

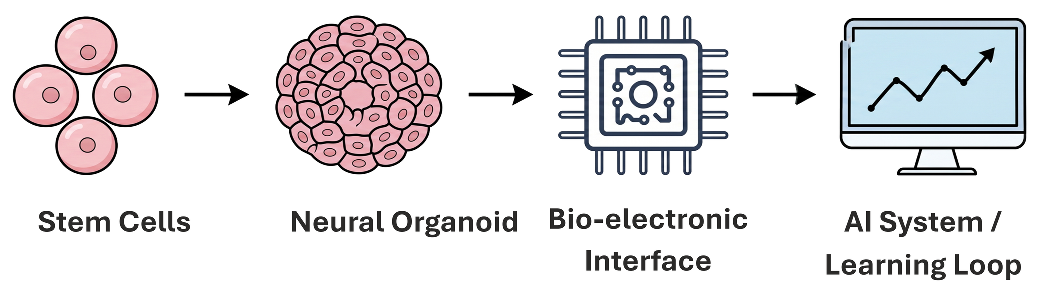
BACKGROUND

Recent advances in neuroscience, synthetic biology and artificial intelligence are enabling a new era of computation based on living neural tissue — organoid intelligence (OI).
These bio-computing systems represent a potential post-silicon paradigm that combines biological neurons with computational interfaces capable of adaptive learning and complex signal processing.

While OI holds promise for medicine, scientific research and energy-efficient computing, it also introduces unprecedented challenges for global security and technology governance.

WHAT IS ORGANOID INTELLIGENCE?

Organoid intelligence uses laboratory-grown clusters of neurons, called organoids, connected to computational systems.
These hybrid bio-digital systems can receive inputs, process signals, and exhibit learning-like behaviors.



Key features: biological adaptability, low energy potential, parallel processing, and emergent dynamics that differ from traditional silicon-based AI.

WHY IT MATTERS FOR GLOBAL SECURITY

AI technology is advancing faster than legislation can keep up, creating regulatory gaps that pose serious ethical and strategic risks.
Organoid AI systems could be leveraged in security-relevant applications such as decision support, autonomous platforms and strategic simulations, raising dual-use and proliferation concerns.
The biological substrate of OI introduces uncertainties related to predictability, control and ethical oversight.
Uncoordinated development may lead to governance gaps, potential misuse and regional instability.

A structured, responsible and inclusive governance approach is essential to ensure responsible innovation and international security.

SECURITY IMPLICATIONS

MILITARY DECISION SUPPORT
Potential use in strategic analysis, battlefield simulations and autonomous systems.

BIOSECURITY RISKS
Dual-use potential and risks of misuse or unauthorized access to biological systems.

ETHICAL AND NEUROETHICAL CONCERNS
Issues related to consciousness, suffering, identity and moral status of synthetic neural systems.

STRATEGIC COMPETITION
Emerging technology race may deepen inequalities and destabilize strategic balances.

UNCERTAINTY AND CONTROL
Unpredictable emergent behaviors and challenges in verification, validation and long-term control.

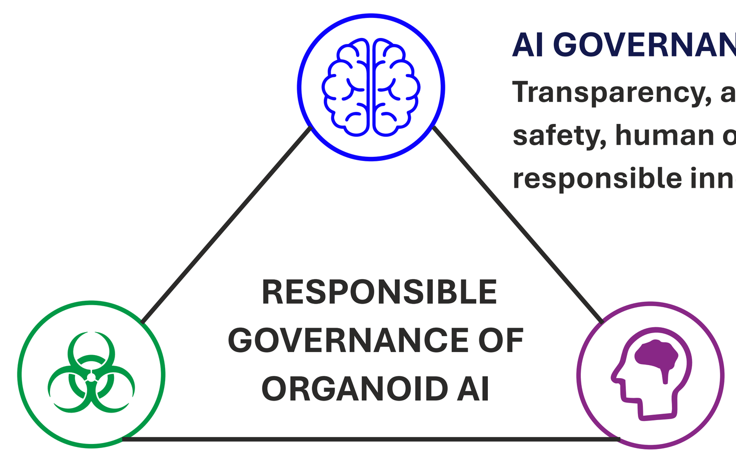
RISK MATRIX

Risk Category	Impact	Likelihood
Dual-use / Proliferation	High	High
Biosecurity Threats	High	Medium
Ethical / Neuroethical	High	Medium-High
Uncertainty / Control	Medium-High	High
Strategic Competition	High	High

Without governance, risks may outpace benefits and create long-term security vulnerabilities.

PROPOSED GOVERNANCE FRAMEWORK

An interdisciplinary framework integrating three key domains is essential for responsible governance of organoid intelligence.



AI GOVERNANCE
Transparency, accountability, safety, human oversight and responsible innovation.

BIOSECURITY
Biosafety, dual-use mitigation, control of bio-computing technologies.

NEUROETHICS
Ethical principles, human dignity, responsible innovation and societal values.

Effective governance requires coordination among states, international organizations, academia, industry and civil society.

POLICY RECOMMENDATIONS

1 Establish International Monitoring Mechanisms
Create an international monitoring system to track advances in organoid AI, share information on bio, computing capabilities, and promote transparency.

2 Develop Biosecurity Standards
Develop biosecurity standards and risk assessment frameworks for organoid AI systems, including guidelines for safe research and laboratory practices.

3 Ensure Transparency and Auditing
Ensure transparency, auditability and bias mitigation through inclusive governance mechanisms and independent oversight.

4 Promote Multistakeholder Engagement
Engage governments, researchers, industry and civil society in developing ethical norms and governance principles.

5 Advance International Cooperation
Strengthen international cooperation to prevent dual-use misuse and ensure responsible, equitable and peaceful innovation.

KEY TAKEAWAYS

- Organoid intelligence represents a potential post-silicon shift in AI.
- Its bio-digital nature introduces new security, ethical and governance challenges.
- An interdisciplinary, anticipatory governance framework is essential.
- International cooperation and inclusive norms can ensure responsible innovation and global security.

LOOKING FORWARD

The governance of organoid intelligence will shape the future of artificial intelligence and international security. Through proactive policies, shared norms and scientific responsibility, the global community can harness the benefits of this emerging technology while minimizing its risks to humanity and peace.

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